Deliverable D - Technical Writeup

Team: the-prior-distribution

1. Problem framing & assumptions violated

We are a small-business lender choosing whom to fund to maximize portfolio NPV, not to maximize classification accuracy. Three properties of the data break the standard i.i.d. supervised-learning assumption and shape every design choice.

Selective labels (the central problem). Repayment outcomes are observed only for loans the prior underwriter approved and that matured: of 85,340 training rows only 51,722 are labeled, and the 33,618 declined applications have no outcome. We must decide on the full validation+test population (13,306 applicants), so the estimand `P(default | x)` is not the labeled distribution `P(default | x, approved)`. Critically, selection is on observables. The approval propensity `e(x)=P(approved | x)` is predicted with AUC 1.000 when the prior underwriter's own score is included, and 0.720 from genuine applicant features alone. We read the 1.000 as near-tautological: the prior score is the decision's basis, not evidence that approval is deterministic in applicant features.

Conditioning on the recorded `prior_underwriter_score` therefore identifies the outcome model where overlap exists, under an ignorability assumption made explicit below. But conditioning on the score relocates rather than removes the positivity problem. Approval is a near-deterministic threshold on that score, so in the feature space the A/B model conditions on, about 43% of the decision set lies in a thin-support prior-score region with little labeled approved analog (`e(x)<0.05`). No reweighting fixes this. We do not decline the whole region by construction: the E[NPV] rule approves where supported features justify positive value and declines the rest, so the book approves about two-thirds of applicants overall. Dropping the score would restore positivity but reintroduce it as omitted confounding, since the underwriter's signal would then be unobserved. We therefore state ignorability-given-the-score as an assumption (the underwriter may also have used private signal), stated explicitly in §3, not as a fact proven by the propensity AUC.

Right-censoring / timing. Default is defined over a 90-day window (3 consecutive missed ACH draws, 6 cumulative misses, or positive balance at day 90) even though the term is 60 days. Median `days_to_default` is 36 and about 21% of defaults occur after day 60, so a binary day-90 classifier discards financially decisive timing information. This forces a survival formulation (§2) shared by Deliverables A and B.

Informative missingness. Bank-feed fields are null exactly when no feed is linked (about 36% of applicants); this is signal, not noise, so we encode missingness indicators rather than impute it away.

2. Methodology

One survival backbone for A and B. We fit a weekly discrete-time hazard `h_a(x)=P(default in week a | survived, x)` over 13 weeks (the 90-day window) on person-period expansions of the labeled loans, using a gradient-boosted classifier with `age_week` as a feature. Survival is `S(t|x)=prod_{k<=t}(1-h_k)`, default probability is `PD = 1-S(13 weeks)`, and the cumulative-default curve feeds B directly. We chose this backbone because it serves A, B, and C coherently. A pre-registered bake-off on an internal train time-split (earlier weeks fit, later weeks select; official validation reserved for calibration) compared six learners. The incumbent histogram-GBM hazard was retained (calibrated Brier 0.1344; the best challenger, CatBoost, was only 0.3% better, below our 2% swap threshold). A calibrated binary logistic PD90 scored within about 1% Brier of the backbone and serves as our simple-model leakage check: it is not anomalously stronger, so the signal is largely additive. We also ran the textbook credit-risk baseline, a pooled logit / complementary-log-log discrete-time hazard on the same person-period expansion. It is marginally better

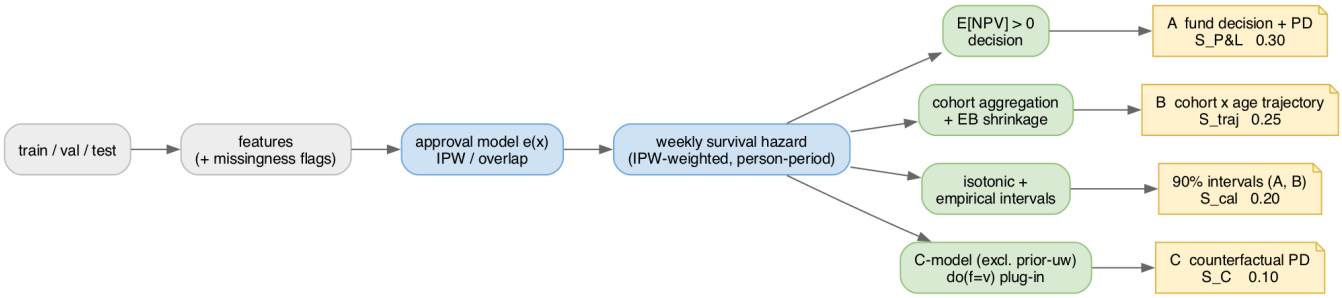


Figure 1: **Figure 1.** One weekly survival backbone (IPW as overlap diagnostic), built around the five scored components: the E[NPV] rule sets funding (S_P&L, 0.30); the same hazard forecasts the cohort trajectory (S_traj, 0.25); isotonic calibration sets the 90% intervals (S_cal, 0.20); a prior-underwriter-free do(f=v) plug-in answers the counterfactuals (S_C, 0.10).

calibrated (cloglog cal-Brier 0.13288 vs the GBM's 0.13437), but by about 1% relative, below our 2% swap bar. We kept the HistGBM hazard for its native missingness handling, and retained the GLM hazard as a calibrated reference. Features: log1p money fields, missingness indicators, GBM-native NaN handling; `prior_decision` and `prior_approved_amount` never enter the risk model.

Selection correction. We fit `e(x)` on all 85,340 rows and form stabilized, clipped inverse-propensity weights for the labeled sample. Because approval is nearly deterministic in the prior score, the estimated propensity is about 1 on every labeled (approved) loan, so the weights collapse to a near-constant: IPW acts as a support/overlap diagnostic, not a de-biaser. The selection correction rests on conditioning, and is limited in the never-approved tail. We verified the risk model is not merely cloning prior policy: dropping the entire prior-underwriter group costs about 0 AUC (0.773 vs 0.774).

Economics and decision. We do not use a stylized accrual formula, because the cure mechanism means collected draws differ from (t-1) draws. Instead we compute realized NPV from the outcome columns and learn loss-given-default empirically by default week: a week-1 default realizes about -0.69 R, a near-day-90 default about +0.09 R, and full repayment about +0.0875 R. Then `E[NPV_i] = S(90)*0.0875*R_i + sum_a P(week a)*LGD_a*R_i`, and we approve iff E[NPV_i] > 0. On realized validation profit this parameter-free policy earns \$2.62M and beats approve-all (\$0.56M) by +370%. It also beats every plausible fixed PD threshold, the policy the brief names as wrong ($p<0.20$ gives \$2.40M, $p<0.30$ gives \$2.53M), landing within 1.4% of the best oracle-tuned threshold ($p<0.24$, \$2.66M) that a flat policy could only find with the unseen outcomes. That gap is the brief's intended insight: low PD is not profitability, and a size-aware NPV rule needs no tuning. All figures are on the labeled (supported) region, the honest caveat. In the thin-support low-score region the rule funds only where E[NPV]>0 and declines the rest (a conservative NPV floor of 0 beats an unsupported approval); this is the standard reject-inference posture for selective labels.

Deliverable B. For each origination cohort week we average `1-S(7a|x)` over our approved cohort-`w` loans, stabilized by empirical-Bayes shrinkage toward the pooled approved-book curve (strength kappa=50) so thin cohorts borrow strength rather than chase noise. Here the conservative book still leaves every cohort well-populated (>550 approved loans each), so the shrinkage is light. We enforce non-decreasing cumulative rates by a cumulative-max pass on the point and both interval bounds; intervals combine model uncertainty with cohort-size sampling variance. On held-out validation cohorts the forecast tracks realized cumulative rates within about 0.7 points per age week (5-fold trajectory MAE about 0.017), and the week-13 cohort default rate is about 14%, consistent with the approved book being lower-risk than the marginal population.

3. Causal reasoning & counterfactual methodology

Deliverable C asks `P(Y | do(f=v), X_{-q}=x)`, an individual estimand that holds the other *free* covariates at their values (deterministic descendants of `f` are not free and move with the intervention). As written this is a per-unit controlled direct effect, not a total effect: any true mediating path from `f` through another observed covariate is held fixed. Under conditional ignorability and overlap, conditioning on `X_{-q}` blocks the backdoor paths, so the identified estimator is the cross-fitted outcome-model plug-in with deterministic descendants recomputed, evaluating the PD model at the intervened covariate vector. This is why we submit the plug-in rather than DML or a causal forest, which target a population-average effect and serve only as cross-checks.

The plug-in is not naive feature perturbation, though the two share the same final arithmetic. Our estimator (i) **recomputes deterministic descendants** of the intervened feature (e.g. the requested-amount-to-revenue ratio moves when `requested_amount` is set; leaving it stale discards the effect that flows through it); (ii) **excludes `prior_underwriter_score` from the C model**, because that score is a descendant of the application features (the underwriter scored them), so freezing it under `do(f)` is the same class of error as not recomputing a ratio, and the audit shows excluding it costs about 0 accuracy; (iii) **audits local overlap** and widens intervals off-support, addressing positivity / thin support; and (iv) **states the unobserved-confounding assumption explicitly** as the key untested assumption. That is a distinct threat from positivity which an overlap band does not bound; a formal E-value sweep is the natural next step, not run here.

We validated this on a semi-synthetic system with a known effect (default risk driven by the leverage ratio, plus a small direct effect; labels made selective by an approval rule). Estimating the dose-response of `do(requested_amount)`:

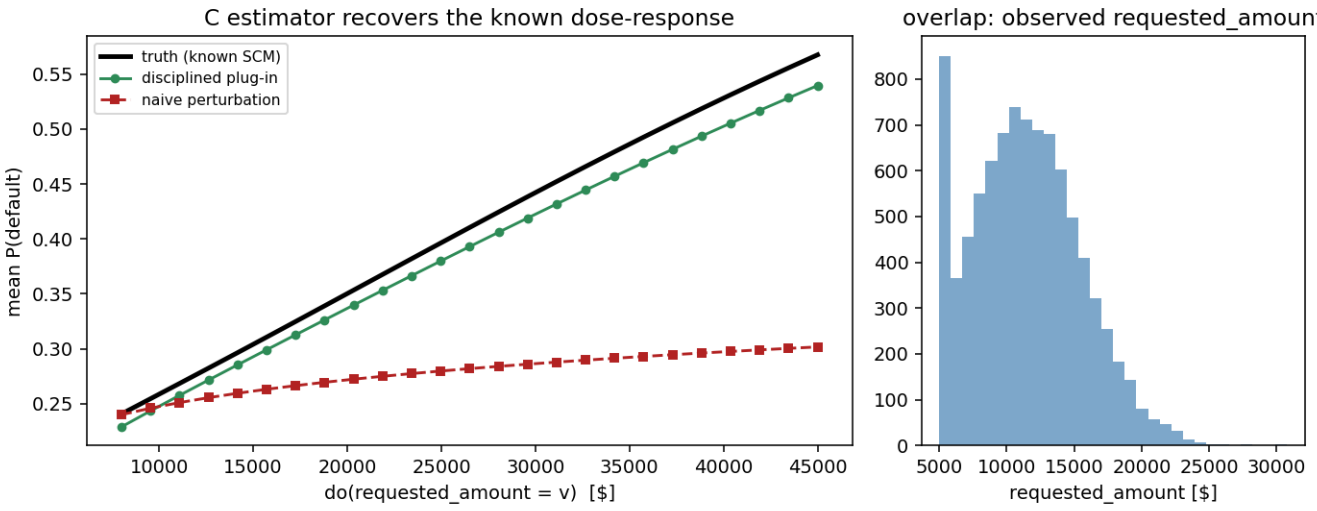


Figure 2: **Figure 2.** Semi-synthetic dose-response under do(requested_amount): the disciplined plug-in (green) tracks the known truth (black) while naive perturbation (red) stays flat. Right panel: observed support. Plug-in MAE 0.018 vs naive 0.130.

The disciplined plug-in recovers the true slope within 6% (+0.084 vs +0.089 per $10k) and is 7.2x closer to truth in mean absolute error than naive perturbation, which underestimates the effect 5.8x because it leaves the stale ratio. This is the difference between observational association and an interventional prediction. On the real data, grouping the 900 queries by feature, a cross-fitted doubly-robust average-effect estimator (partial-linear DML, prior-underwriter excluded, approval IPW) agrees in sign with our plug-in on all 5 features the model materially responds to. We claim sign agreement only: the magnitudes differ (e.g. credit

utilization +0.12 plug-in vs +0.29 DML), which is expected because DML targets a population-average effect while the plug-in is an individual conditional one. They are different estimands, so we do not read DML as validating our magnitude. The submitted per-query value is the conditional plug-in by design.

Defending drivers to a regulator. Our stated drivers (leverage, credit utilization, delinquency, cash-balance signals; see the selection and risk audits) are conditioned-on observables with stable, signed, monotone relationships to default (e.g. default rate falls monotonically across prior-score deciles, 0.44 to 0.14). We exclude the prior decision itself to avoid laundering historical policy, report risk with and without the prior-underwriter group, and attach explicit overlap and confounding caveats rather than claiming identified effects off-support.

4. Calibration & uncertainty quantification

Point PDs are calibrated by isotonic regression fit on the labeled validation set, improving cross-fit ECE from 0.026 to 0.018; the same monotone map rescales the survival curve so A and B remain consistent. Because the scored target is the latent PD (a number), A's 90% interval is a tight epistemic confidence interval on the estimate: bootstrap-ensemble spread plus a small floor (mean width about 0.13). We do not inflate it with binary-outcome variance, which would make it needlessly wide. On the late time-split this width achieved 0.900 binned coverage at the minimum tested safe setting; tightening (x0.75) under-covered (0.70), and widening (x1.25) only added width (1.00). For B, the scored object is a cohort fraction, so its 90% interval is a proper combination of ensemble spread and binomial sampling over the `n_w` approved loans in each cohort (mean width about 0.03). C is scored on point accuracy rather than interval calibration, so there we prioritize the plug-in estimate and only widen intervals off-support as a transparency flag. The trade-off is to target about 90% coverage at minimum width where calibration is scored (A, B), and to spend C's budget on point accuracy.

5. Limitations & what we'd do differently

Temporal drift (first and sharpest). The observed default rate rises from 17.4% (train) to 20.6% (the later validation block), and the test window is later still. We calibrate isotonic on validation, the most recent labeled block, so the train-to-validation drift is absorbed; but if the upward trend continues into test, our PDs will under-predict, which hurts P&L (we fund loans riskier than we think) and calibration (intervals miss low) together. 5-fold CV (random and time-blocked) shows the backbone is stable (AUC 0.77 +/- 0.01 both ways, so not overfit) but does not correct a trend: the latest rolling folds are measurably harder (Brier about 0.11 to 0.13). We do not claim the drift is removed; a conservative production hedge would nudge the global PD level toward the most recent observed rate.

Positivity / reject inference. About 43% of the decision set is a thin-support prior-score region with little labeled analog. The E[NPV] rule declines its weakest-support part (the book approves about two-thirds overall), not as profit-maximizing but as declining where we cannot evaluate. Two further caveats: loss-given-default is noisy (collected draws before default are not directly observed, so our empirical LGD-by-week is an approximation), and C is a controlled direct effect (we treat the remaining covariates as non-descendants of the intervened feature except the deterministic ratio, so if true mediating paths exist, e.g. revenue moving overdrafts, we estimate a direct effect through a partially-blocked path, not a total effect).

Counterfactuals on weak features. The tree plug-in is near-flat for low-importance features, so counterfactuals on weak drivers are conservative; the DML cross-check confirms only the material drivers carry effect. We tested whether this flatness costs accuracy. On a semi-synthetic test-bed injecting a known weak-feature effect, the tree recovered 79% of it. A smoother logistic C-model recovered more on the test-bed but over-extrapolated on the real in-support queries (PD around 0.91 under revenue interventions, an off-manifold extrapolation), further from plausible truth than the bounded tree, so we kept the tree.

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